

AY 2021-2022

LIFE SKILLS (2021-22)	
Physical Fitness	September, 2021/ February, 2022
Yog Vidya	September, 2021/ February, 2022
Jeevan Vidya Work Life Balance	September, 2021/ February, 2022
Innovation and digital Eco system	September, 2021/ February, 2022
Study of any of GREAT sons of INDIA	September, 2021/ February, 2022
Green Initiative	September, 2021/ February, 2022
Study of Green & White Revolutions in India	September, 2021/ February, 2022
Government Missions [Study of any 2]	September, 2021/ February, 2022
Study of India's top 2 problems	September, 2021/ February, 2022
Study of World's top 2 problems	September, 2021/ February, 2022
Study of one of the identified Books	September, 2021/ February, 2022
Study of two National policies	September, 2021/ February, 2022
Study of Corona Pandemic in Maharashtra or India	September, 2021/ February, 2022
Mentoring of School Children	September, 2021/ February, 2022



Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai)

[Knowledge is Nectar]

Liberal, Pi-Model of Engineering Education @ SPIT **(Department of Computer Engineering)**

CURRICULUM SCHEME FOR UNDERGRADUATE ACADEMIC PROGRAM (COMPUTER ENGG) AT SPIT

(For 2021-2025 Batch)

Salient Features

- 160-Credit **Liberal** Engineering Education Model.
- A strong **program core of 13 courses** and **6 baskets of program electives** to ensure the breadth and depth in a chosen domain of studies. Program electives are arranged either to grow in a specified vertical or have diversified exposure.
- **Full semester industry internship to interested students.**
- Aggressive model of “**Learning-by-doing**”. (Engagement in classroom and laboratory sessions is 50:50)
- Special tracks for “**Minor**” Certification for interested learners, ensuring significant awareness of additional discipline leading to multiple specializations
- **Unique, multi-track model of “Honors” Certification**, for well performers for enhanced depth in the domain of study.
- Special sequel of optional **industry floated “SCOPE”** courses (Skilled Certification for Outcome-based Professional Education) for interested learners, ensuring high technical skills, in the diversified cutting-edge technologies.
- **First-of-its-kind-in-education** blend to Engineering Curriculum. “**ABLL@LLC**”[®] (Activity Based Liberal Learning about **Life, Literature and Culture**) in **Six** semesters, ensuring **all dimensional holistic growth** of the learner. These eight activity based mini courses are offered as two sequels namely “**SEVA**”[®] (Social Empowerment through Various Activities), and “**SATVA**”[®] (Self accomplishment through various Activities).

This curriculum aims at development of an **all-rounded** personality. It follows **holistic** approach of education, ensures strong science, mathematics foundation and program core, develops expertise in domain vertical though sequel of electives, ensures significant exposure of additional discipline through “Minor” program, collaborates outside world for the imparting relevant skills through “SCOPE” courses, challenges good learners through “Honors” evaluation, and systematically develops soft skills, and social,

physical, mental, spiritual personality through carefully articulated **Liberal Learning** and **Humanities** sequels. Thus, offers a unique, liberal **“Pi-Model”** of Engineering Education.

Program Core

At SPIT, every undergraduate program consists of **Twelve Core Courses** referred to as Program Core. Several academic models from reputed institutions in the country and outside the country are studied in articulating this Program Core, to make curriculum Globally Competitive. All courses in this Core have laboratory components to augment the learning. Each program core course has an additional optional component of “Contents beyond the curriculum” which is carefully designed to ensure additional 15-20 hours engagement of the learners. The learner thus is nurtured towards the “Self-Learning” and “lifelong learning” which are essential attributes of 21st Century learner.

Program Electives

At SPIT, every program has **Six baskets** of Program Electives, each basket having a minimum 3 courses. This enables learners to grow in a **domain-specialization** or **domain-vertical**. For example, learners can graduate with B.Tech Electronics with a vertical in “Embedded Systems” or “VLSI” or “Signal Processing”. Or a learner can graduate with B.Tech Computer Engineering with specialization in “Security” or “ML & AI” or “Computer Networking” or “Data Science”. At the same time, a learner can increase her bandwidth by opting for elective courses which are general in nature, not pointing out towards a specific vertical.

Open Electives

Every undergraduate program has three baskets of open electives. This is planned to give exposure to interdisciplinary and cross disciplinary domains. The courses in these baskets are planned both at department and institute level. Students can choose any combination of these courses (not floated by the parent department) to get familiar with other domains of learning. One of these open electives must be chosen from Basic science courses or Engineering Science courses. **This unique approach of offering additional basic science or engineering science elective at senior level aims at appreciating the importance of other domains of learning.**

Humanities and Social Science Electives

National Education policy 2019 has aptly spelled out the necessity of Humanities in Professional Education. It quotes, “A holistic and liberal education as described so beautifully in India’s past is indeed what is needed for the education of India in the future to truly lead the country into the 21st century and the fourth industrial revolution. Even engineering schools such as the IITs must move towards a more liberal education integrating arts and humanities”. Every program at SPIT has three baskets of humanities. Learners are encouraged to take diversified courses in the field of languages, law, history, economics, management, finance etc.

SCOPE Certification

This unique sequel is designed to systematically develop skills required for an industrial sector. SPIT is partnering with various industries to offer the high-end skills required for a specific

industrial sector. Well performing students can stretch the envelope and add a new dimension to their Professional Personality by earning this certification. There are multiple tracks for SCOPE certification. Each track is offered with partnership with a reputed institution or industry. These tracks are jointly designed by SPIT and partnering industry. Each track has four courses (modules). Each module/course is of 2-3 credits including laboratory components for most of the tracks. These tracks are also open for outside learners, leading to Certificate Program in a chosen domain.

Minor Certification

This additional and optional certification provides an opportunity to learners to develop the learners in the additional domain of interests. It broadens the education and ensures the multi-disciplinary development which is an essential attribute of 21st century engineers. However, this is optional. Well performing students can stretch the envelope and add a new dimension to their Professional Personality. Each track for this minor certification is offered either by SPIT or with partnership with other reputed institutions. Each track has four courses (modules). Each course is of 3 credits and laboratory components if any. These tracks are also open for outside learners, leading to a Certificate Program of 12 credits in a chosen domain.

Honors Certification

While the Minor and SCOPE certifications aim at adding an additional professional dimension to the professional personality of the learners, the Honors certification gives opportunity to well performing learners to drive deep in the chosen field of study. Multiple plans/ways are planned to encourage learners to earn this certification which essentially excite the learners to push an envelope and go extra/deep in the chosen area of the study. Students earn additional stars (*) as shown in Table 1 during their program. If at the time of graduation a student earns total **TWELVE** stars, she is conferred with "Honors" certification.

ABLL@LLC

- Students are required to earn 6 credits through 8 semesters.
- If student is not able attendance/performance requirements, he/she will be dropped from the course and will have to enroll in additional course in the next semester.
- A student can enroll in maximum 2 courses in a semester.

SEVA (Social Empowerment through Various Activities)			
Module	Title	Courses	CODE
SEVA-I	SOCHO BHARAT	Study of Green & White Revolutions in India	SV 10
		Government Missions [Study of any 2]	SV 11
		Study of India's top 2 problems	SV 12
		Study of World's top 2 problems	SV 13
		How Government Works? [Study of one department of the Central/ State Government]	SV 14
		Study of one of the identified Books	SV 15
		Study of two National policies	SV 16
		Any other activity approved by Dean Academics	SV1X
Module	Title	Courses	CODE
SEVA-II	SWACCH BHARAT	River/Beach/Mohalla/School/Campus/Govt offices Cleaning	SV 20
		Waste Segregation Surveys	SV 21
		NSS camp in village for a week	SV 22
		Medical camps in schools	SV 23
		First Aid training for a week	SV 24
		Surveys and Estimation for roof top solar	SV 25
		NCC participation	SV 26
		Any activity approved by Dean Academics	SV2X
Module	Title	Courses	CODE
SEVA-III	SHIKSHIT BHARAT	Mentoring of School Children	SV 30
		Digital Literacy for yielders	SV 31
		Value addition for deprived schools	SV 32
		Mentoring junior (first year) students at SPIT	SV 33
		Teaching Assistantship at SPIT	SV 34
		Development of learning material for schools/ITIs	SV 35
		Participation in "Teach-for-India" movement	SV 36
		Any other activity approved by Dean Academics	SV3X
Module	Title	Courses	CODE
SEVA-IV	SAMRUDDHA BHARAT	Great Grass Root Innovations	SV 40
		Innovation and Creativity	SV 41
		Critical Thinking and Problem solving	SV 42
		Team work and collaboration	SV 43
		Leadership & Entrepreneurship	SV 44
		Design Thinking	SV 45
		Study of one of the identified books	SV 47
		Work with START-UP at SPIT	SV 48
		Any other activity approved by Dean Academics	SV 49

SATVA (Self Accomplishment Through Various Activities)			
Module	Title	Courses	CODE
SATVA-I	SANSKRIT BHARAT	Values and Ethos of Bhavan	ST10
		Essence of Indian traditional knowledge	ST11
		Philosophy of religion (any)	ST12
		Study of Life Management / Kindle Life / Life Empowerment and Enriching Program or any other book cited.	ST13
		Study of any of GREAT sons of INDIA [Ex. Gandhi, Ambedkar, Phule, Savarkar, Sardar Patel, Nehru, Shivaji, JRD Tata etc]	ST14
		Any other course approved by Dean Academics	ST1X
SATVA-II	SAKSHAM BHARAT	Target based Physical Exercise for example-Running [Test 5 kms in a stretch], Swimming [Test 1 km in a stretch], Walking [Test 20 kms in a stretch], Trekking [7days], Cycling	ST20
		Sports – Representation of Institute at University level/Inter college level and above in ANY sport	ST21
		Participation in National Tech Fest, AICTE-Hackathon, Industry floated global and national competitions, Robocon, BAHA etc	ST22
		Yoga vidya -I	ST23
		Any other activity approved by Dean Academics	ST2X
SATVA-III	SUNDER BHARAT	Institute representation in prestigious cultural fests/competitions	ST30
		Dance [Bharatanatyam /Kathak /Lavani /Western Dance]. Only for beginners	ST31
		Learning musical instruments [Any type]. Only for beginners.	ST32
		Film Appreciation/Dramatics/Seeing through Painting	ST33
		Making short film/Photography	ST34
		Yogvidya-II	ST35
		Any other activity approved by Dean Academics and DOSA	ST3X
SATVA-IV	SURAKSHIT BHARAT	Food that Heals	ST40
		Personal and Social Hygiene	ST41
		Intellectual Property Rights	ST42
		Etiquette and Conversational skills	ST43
		Races of Ayurveda	ST44

Sem II									
No.	Type	Code	Course	L	T	P	O	E	C
1	BSC	MA102	Differential Equations and Complex Analysis	3	1	0	8	12	4
2	BSC	AS101	Engineering Physics	2	1	2	5	10	4
3	ESC	AS104	Engineering Graphics	1	0	2	2	05	2
4	ESC	ET101	Basic Electrical Engineering	3	0	2	6	11	4
5	ESC	CS102	Problem Solving using OOP	2	0	4	6	12	4
6	SBC	AS106	Skill Shop	0	0	2	0	02	1
7	ABL	SV1X/ST1X	SEVA-I or SATVA-I	0	0	0	2	02	1
TOTAL				11	2	12	26	51	20

Sem III									
No.	Type	Code	Course	L	T	P	O	E	C
1	BSC	MA203	Probability and Statistics	3	0	0	5	08	3
1	BSC*	MA202	Foundation of Mathematics-I*	2	1	0	6	09	3
2	PC	CS201	Discrete Structures and Graph Theory	3	0	0	4	07	3
3	PC	CS202	Data Structures	3	0	2	5	10	4
4	PC	CS203	Computer Architecture and Organization	3	0	2	4	09	4
5	PC	CS204	Database Management Systems	3	0	2	5	10	4
6	ABL	SVXX/STXX	SEVA II or III /SATVA II or III	0	0	0	3	03	1
7	HSSE	HSEX1	HSS-I	2	0	0	3	05	2
TOTAL				17	0	6	29	52	21

**Only for Lateral Entry Students*

Sem IV									
No.	Type	Code	Course	L	T	P	O	E	C
1	BSC	MA201	Linear Algebra	2	0	2	5	09	3
1	BSC*	MA204	Foundation of Mathematics-II	2	1	0	6	09	3
2	PC	CS205	Design and Analysis of Algorithms	3	0	2	5	10	4
3	PC	CS206	Operating Systems	3	0	2	5	10	4
4	PC	CS207	Computer Communications and Networks	3	0	2	5	10	4
5	SBC	CS208	Mini Project-I	0	0	0	4	04	2
6	ABL	VXX/STXX	SEVA II or III /SATVA II or III	0	0	0	3	01	1
7	HSSE	HSEX2	HSS-II	2	0	0	3	05	2
8	SBC	AS201	Professional Communication Skills	1	0	2	2	05	2
9	S/M	SCX1/MNX1	SCOPE-I/Minor-I						3
TOTAL				14	0	10	32	56	22

**Only for Lateral Entry Students*

Sem V									
No	Type	Code	Course	L	T	P	O	E	C
1	PC	CS301	Theory of Computation	3	0	0	6	9	3
2	PC	CS302	Software Engineering	3	0	2	5	10	4
3	PC	CS303	Artificial Intelligence and Machine Learning	3	0	2	5	10	4
4	PC	CS304	Distributed Computing	3	0	2	5	10	4
5	SBC	CS305	Cloud and Internet Technology Lab	1	0	2	5	08	2
6	HSSE	HSEX3	HSS-III	2	0	0	3	05	2
7	ABL	SVXX/STXX	SEVA II or III /SATVA II or III	0	0	0	3	3	1
8	S/M	SCX2/MNX2	SCOPE-II/Minor-II						3
TOTAL				15	0	8	29	52	20

Sem VI (Cat 1- For Students who have NOT preferred semester long internship)									
No	Type	Code	Course	L	T	P	O	E	C
1	OE	OEXXX	Open Elective-I	2	0	2	4	8	3
2	PC	CS306	System Programming and Compiler Construction	3	0	2	5	10	4
3	PC	CS307	Fundamental of Signal and Image Processing	3	0	2	5	10	4
4	PE	CS3X1	PE-I	2	0	2	4	8	3
5	PE	CS3X2	PE-II	2	0	2	4	8	3
6	SBC	CS308	Main Project Stage-I	0	0	0	8	08	3
7	ABL	SVXX/STXX	SEVA II or III /SATVA II or III	0	0	0	3	3	1
8	S/M	SCX3/MNX3	SCOPE-III/Minor-III						3
TOTAL				6		4	18	28	21



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India

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Mentoring of School Children

September, 2021/ February, 2022

A handwritten signature in blue ink, likely belonging to the Dean Academics.

Dean Academics





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SEVA / SATVA

STUDY OF GREEN &
WHITE REVOLUTIONS
IN INDIA

SEPTEMBER, 2021/
FEBRUARY, 2022



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SEVA/SATVA

Yog Vidya

By Bhavan's yoga

Date: September, 2021/ February, 2022



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STUDY OF WORLD'S TOP 2 PROBLEMS

SEPTEMBER, 2021/
FEBRUARY, 2022



DATE: 23-01-2023 TO 05-05-2023

VENUE: SARDAR PATEL INSTITUTE OF
TECHNOLOGY

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CRITICAL READING

22 JANUARY
2021

Pack your enthusiasm
and curiosity!



Alister

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INNOVATION AND CREATIVITY

Let's explore technology
together to live in the future



September 2021/ February, 2022

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JEEVAN VIDYA

DISCOVER THE ART OF LIVING WITH JEEVAN VIDYA

SEPTEMBER, 2021/ FEBRUARY, 2022

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SEVA SATVA

PHYSICAL FITNESS

DATE : SEPTEMBER 2021/ FEBRUARY, 2022



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India
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Study of Corona Pandemic in Maharashtra or India

September, 2021/ February, 2022



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Study of India's top 2 problems

September, 2021/ February, 2022

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Dean Academics



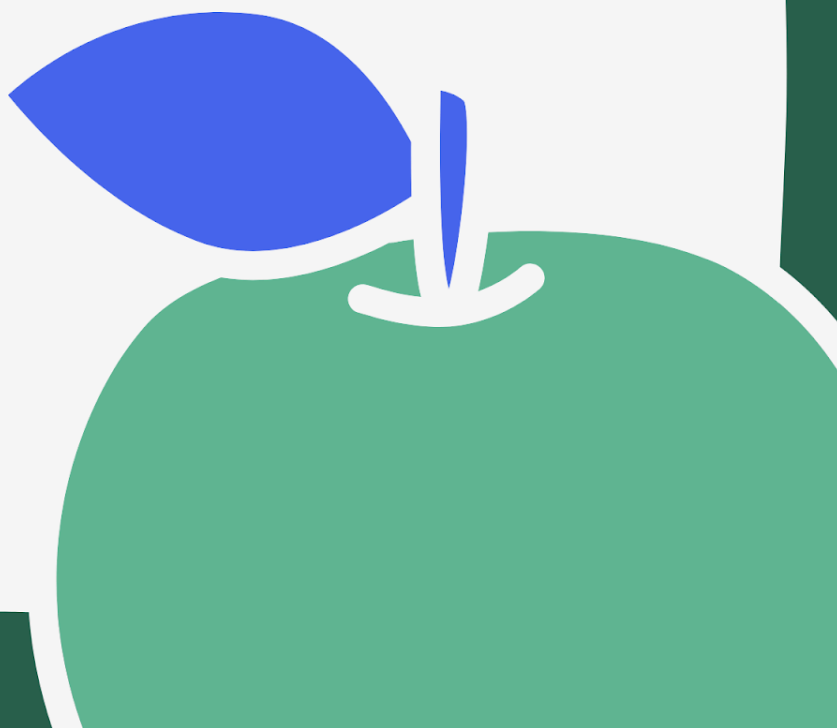


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SEVA/SATVA

GREEN INITIATIVE

September, 2021/ February, 2022



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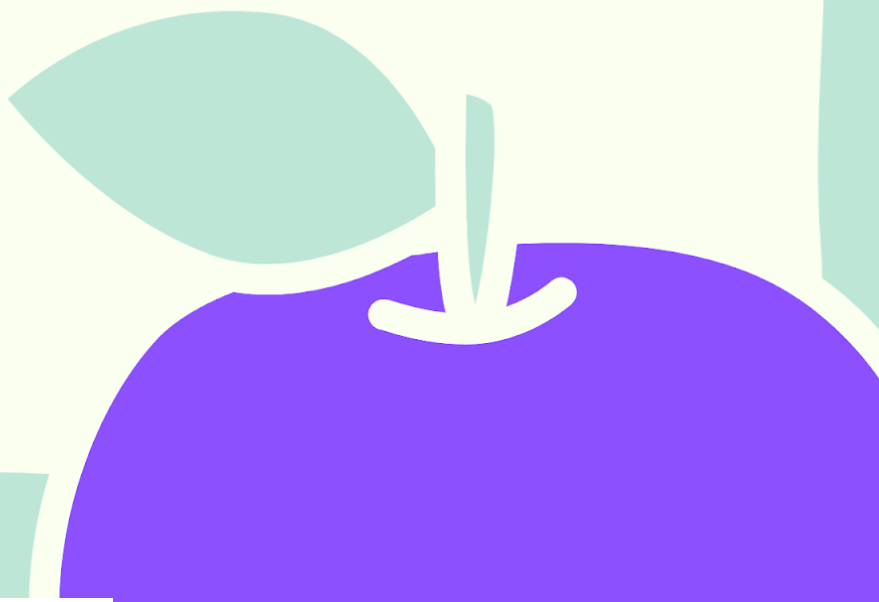
India

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SEVA/SATVA

GOVERNMENT MISSIONS [STUDY OF ANY 2]

September, 2021/ February, 2022



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SEVA/SATVA
**STUDY OF TWO
NATIONAL POLICIES**



SEPTEMBER, 2021/ FEBRUARY, 2022

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